

Introduction to Quantitative Biology

Lecture Notes: Biological Oscillations

From An Introduction to Systems Biology — Uri Alon
Nonlinear Dynamics and Chaos — Steven H. Strogatz

September 2026

1. Biological question: how can a biological oscillator be built?

Oscillations appear repeatedly in biological systems, including the cell cycle, heart rhythms, neuronal activity, and transcription-factor dynamics. A recurring starting point is **negative feedback**: when a variable increases, the circuit eventually acts to reduce it.

However, negative feedback by itself is not sufficient to generate sustained oscillations.

The main question is therefore not simply whether a circuit contains negative feedback, but **what additional dynamical ingredients make oscillation possible and robust**.

2. Why an autonomous one-dimensional system cannot oscillate

Consider the one-dimensional system

$$\frac{dx}{dt} = P(x) - R(x) \equiv F(x),$$

where $P(x)$ is the production rate and $R(x)$ is the removal rate.

At any particular value of x , the sign of $F(x)$ uniquely determines the direction of motion:

$$F(x) > 0 \Rightarrow x \text{ increases}, \quad F(x) < 0 \Rightarrow x \text{ decreases}.$$

A periodic trajectory would have to pass through the same value of x once while moving upward and again while moving downward. An autonomous one-dimensional equation cannot assign two different directions to the same state. Therefore, it cannot generate an oscillation.

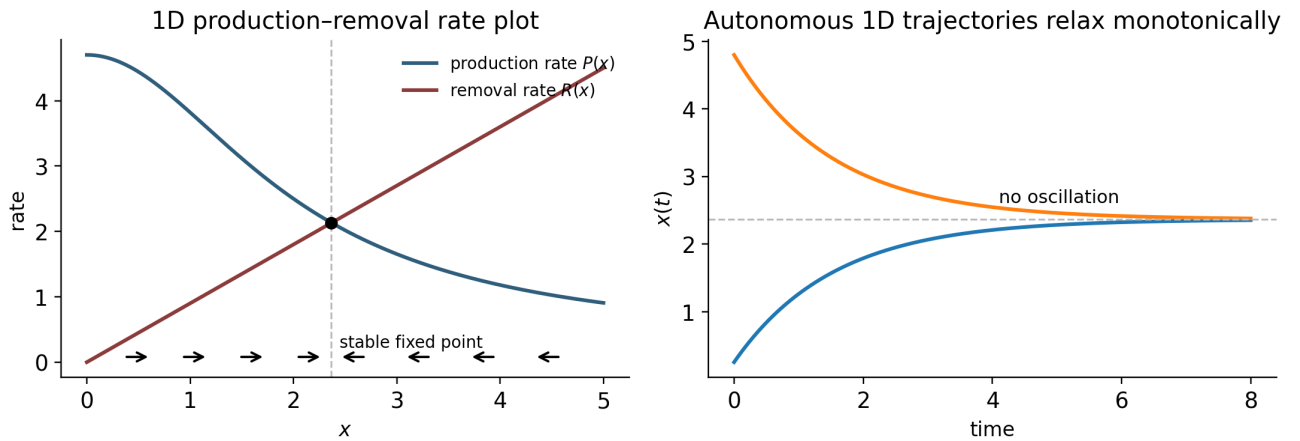


Figure 1. Production-removal rate plot and one-dimensional relaxation. The graphical argument does not depend on the exact form of the production function.

Adding an explicitly time-varying parameter, a delay, or another dynamical variable can change this conclusion, but then the system is no longer an autonomous one-dimensional system.

3. A two-node nonlinear negative-feedback system

Consider a two-node negative-feedback circuit in which X activates Y , while Y represses X . We keep the regulatory input functions nonlinear:

$$\frac{dX}{dt} = \beta_X f(Y) - \alpha_1 X$$

$$\frac{dY}{dt} = \beta_Y g(X) - \alpha_2 Y$$

Here $f(Y)$ is a decreasing repression function and $g(X)$ is an increasing activation function; Hill functions are a typical choice.

The parameters α_1 and α_2 are first-order removal rates and set the inverse timescales of the two arms,

$$\tau_1 \sim \frac{1}{\alpha_1}, \quad \tau_2 \sim \frac{1}{\alpha_2}.$$

Following the notation used for the feedback strength near the fixed point (X^*, Y^*) , define

$$\beta_1 = -\beta_X f'(Y^*) > 0, \quad \beta_2 = \beta_Y g'(X^*) > 0.$$

Thus, β_1 and β_2 describe the local feedback gains, whereas α_1 and α_2 describe removal rates / timescales.

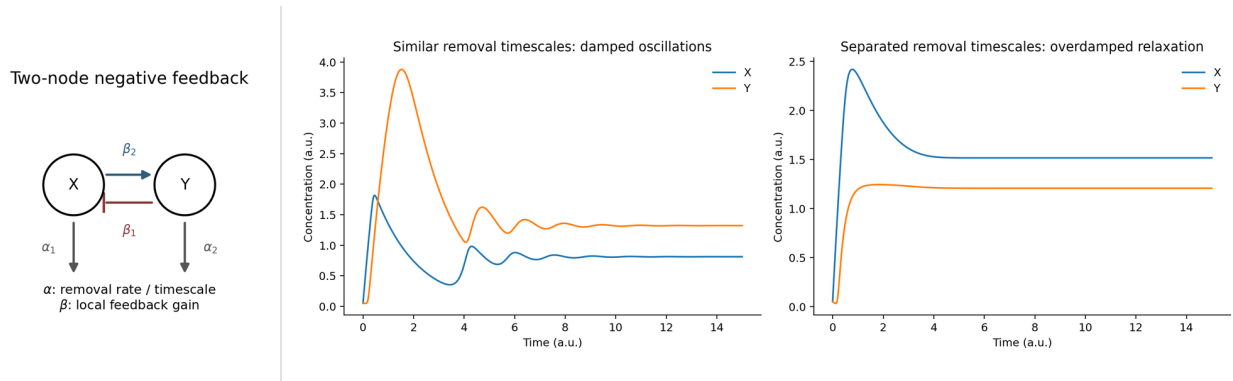


Figure 2. Left: two-node negative feedback. The α_i are removal rates / inverse timescales and the β_i denote local feedback gain. Right: numerical solutions of the full nonlinear equations. Similar removal timescales can give damped oscillatory relaxation, whereas strongly separated removal timescales give an overdamped response.

3.1 Condition for damped oscillations

For the nonlinear two-node system, the local condition for damped oscillations around the fixed point is

$$(\alpha_1 - \alpha_2)^2 < 4\beta_1\beta_2.$$

This result gives the main intuition directly: **damped oscillations are easier when the two removal rates are similar and when the feedback gains β_1 and β_2 are sufficiently strong.**

The curves in Figure 2 are obtained by numerically integrating the full nonlinear equations above; the nonlinear production functions are not replaced by linear production terms.

3.2 Stable spiral versus stable node

The same distinction can be seen in phase space. When the damped-oscillation condition is satisfied, trajectories approach the fixed point as a **stable spiral**. Otherwise, they approach as a **stable node** without oscillatory relaxation.

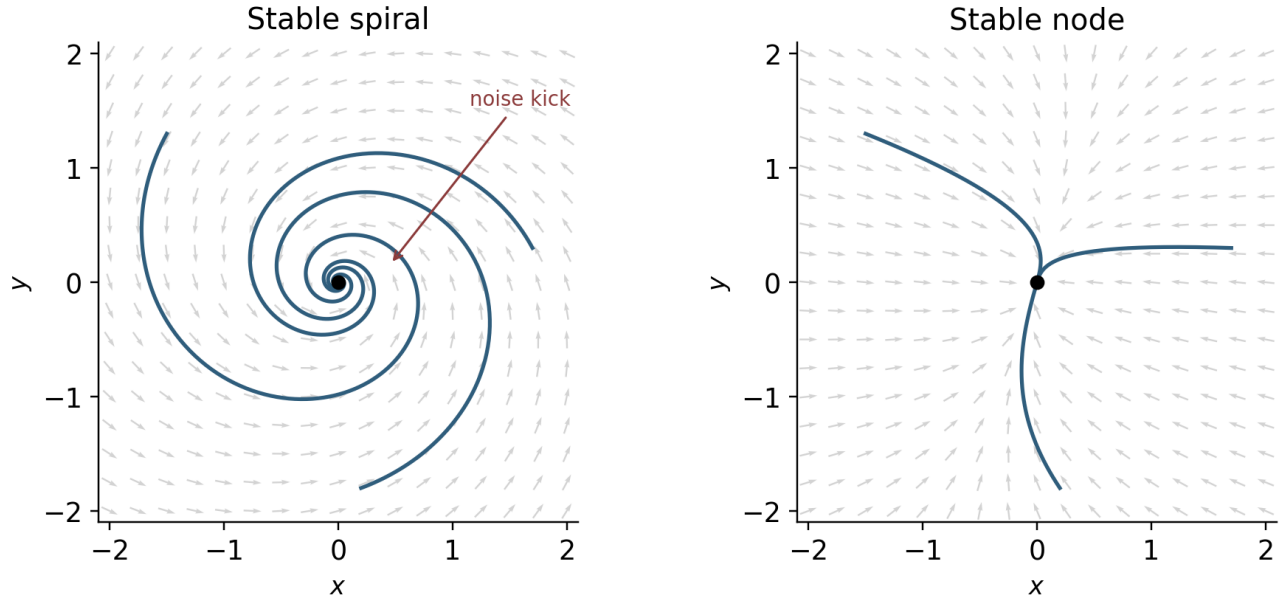


Figure 3. A stable spiral corresponds to damped oscillations; a stable node corresponds to non-oscillatory relaxation. In both cases, trajectories ultimately approach a stable fixed point.

This distinction is important for the next section: noise can repeatedly kick a stable spiral away from the fixed point and thereby sustain oscillatory excursions.

3.3 The two-node system cannot sustain a limit cycle

Although this nonlinear two-node negative-feedback system can show damped oscillations, it cannot sustain a periodic limit cycle. Because $\alpha_1 > 0$ and $\alpha_2 > 0$, the phase-space flow is

contracting.

two-node negative feedback
 $\rightarrow$ damped oscillations are possible, but a sustained limit cycle is not.

Proof: exercise.

4. Noise can sustain oscillations: p53–Mdm2

The p53–Mdm2 circuit provides a biological example of negative feedback. DNA damage activates p53; p53 promotes Mdm2 production; and Mdm2 promotes p53 degradation.

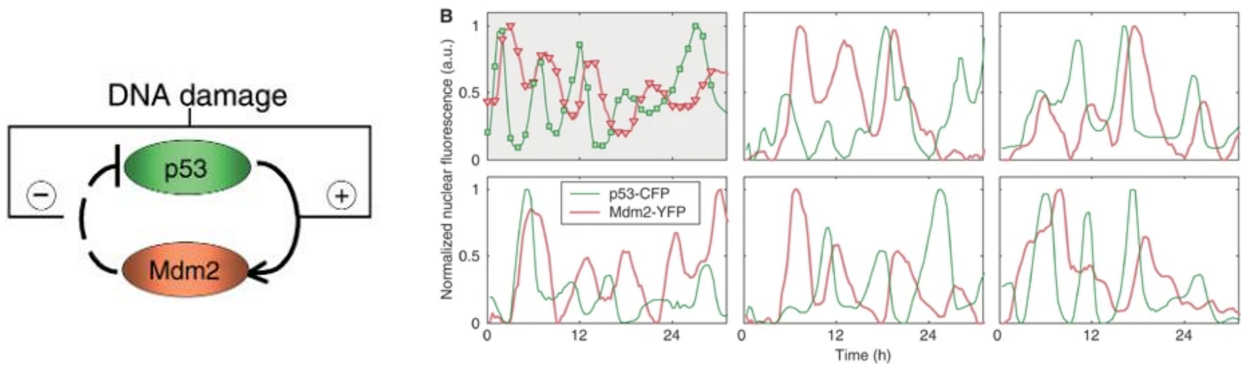


Figure 4. Left: p53–Mdm2 negative-feedback circuit after DNA damage (Lahav et al., *Nature Genetics*, 2004). Right: single-cell p53-CFP (green) and Mdm2-YFP (red) dynamics showing large pulse-to-pulse and cell-to-cell amplitude variability (Geva-Zatorsky et al., *Molecular Systems Biology*, 2006, Fig. 1B).

After DNA damage, individual cells show repeated p53 and Mdm2 pulses. The pulse period is relatively well defined, whereas the pulse amplitude is much more variable.

A deterministic two-node negative-feedback system can show only damped oscillations. In a cell, however, molecular fluctuations can repeatedly perturb the system away from its fixed point before the oscillation has completely decayed. The result is

damped oscillator + noise $\rightarrow$ persistent noisy oscillations.

The underlying oscillator still has a characteristic timescale, which sets the preferred period of these noise-driven excursions.

For one oscillatory variable,

$$T = \text{period}, \quad f = \frac{1}{T}, \quad \omega = \frac{2\pi}{T},$$

and the amplitude measures the excursion away from the baseline or fixed point.

When individual cells oscillate with different phases, averaging many cells can make sustained single-cell pulses appear as a damped population-level response.

5. Amplitude modulation and frequency modulation

Oscillations can also encode the strength of an input. Two simple possibilities are **amplitude modulation (AM)** and **frequency modulation (FM)**.

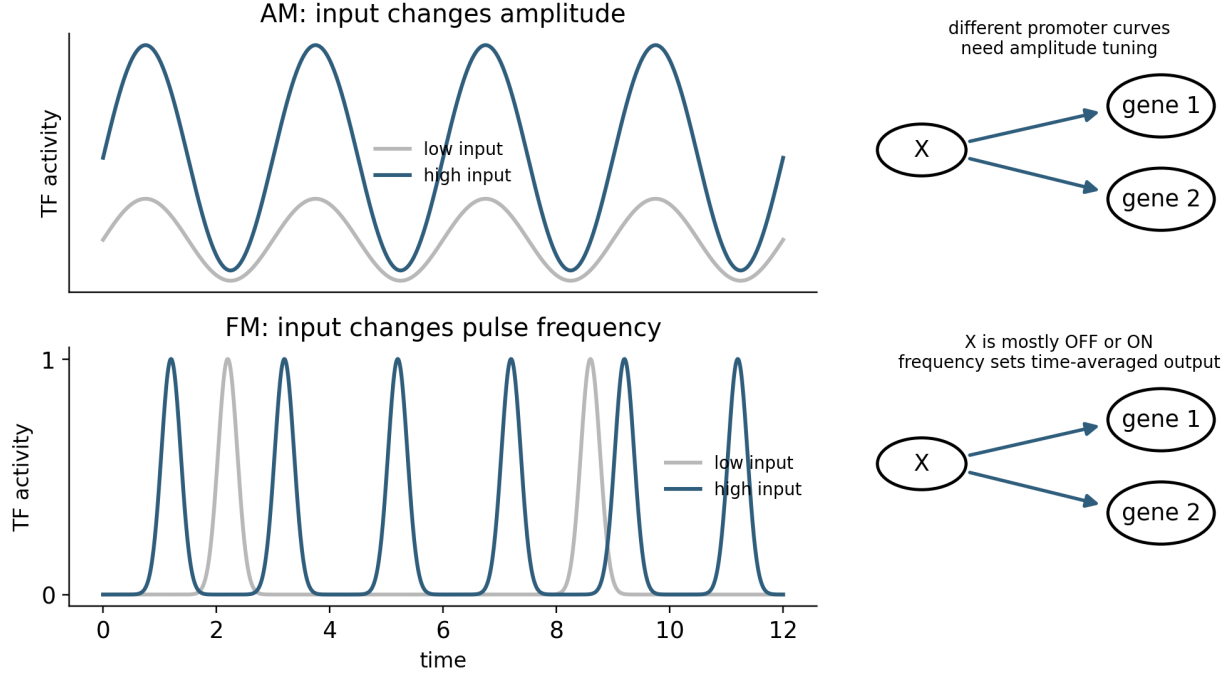


Figure 5. AM changes the size of the response. FM keeps an approximately fixed ON state and changes how frequently the system enters that state.

5.1 AM

In AM, input strength is represented by response amplitude. Suppose two downstream targets respond to the same regulator X as

$$g_1(X) = \beta_1 \frac{X}{K_1 + X}, \quad g_2(X) = \beta_2 \frac{X}{K_2 + X}.$$

Their ratio is

$$\frac{g_1(X)}{g_2(X)} = \frac{\beta_1 \frac{X}{K_1 + X}}{\beta_2 \frac{X}{K_2 + X}}.$$

Rearranging,

$$\frac{g_1(X)}{g_2(X)} = \frac{\beta_1}{\beta_2} \frac{X}{K_1 + X} \frac{K_2 + X}{X}.$$

The factor X cancels:

$$\boxed{\frac{g_1(X)}{g_2(X)} = \frac{\beta_1 K_2 + X}{\beta_2 K_1 + X}}.$$

However, the ratio still contains X . Therefore, when the input level X changes, the ratio g_1/g_2 will generally change as well. If the desired behavior is that a stronger upstream signal simply scales the whole downstream program while keeping the relative proportions of downstream genes unchanged, then AM does not naturally guarantee that property.

5.2 FM

In the simplified FM picture used in class, a regulator is mostly in one of two states,

$$X \approx 0 \quad \text{or} \quad X \approx X_{\text{high}},$$

and stronger input increases the fraction of time spent in the ON state or the number of pulses per unit time.

If the system is ON for a fraction q of the time,

$$\langle g_i \rangle = q g_i(X_{\text{high}}),$$

so

$$\frac{\langle g_1 \rangle}{\langle g_2 \rangle} = \frac{g_1(X_{\text{high}})}{g_2(X_{\text{high}})}.$$

The ratio no longer depends on the duty cycle q . This is the basic **digital / bang-bang** intuition discussed in class: use a robust ON state and regulate how often it is used.

The trade-off is at weak input. FM then means very rare pulses, so the readout can become dominated by the stochastic number of pulses in a finite observation window. In that regime, AM can offer better resolution.

Optional reading — not covered in class

- Yissachar et al. (2013), *Molecular Cell*, “**Dynamic response diversity of NFAT isoforms in individual living cells.**” This is the paper connected to Nir Friedman mentioned in class. NFAT1 and NFAT4 provide an example of different dynamic encoding modes.
- Cai, Dalal & Elowitz (2008), *Nature*, “**Frequency-modulated nuclear localization bursts coordinate gene regulation.**” This paper studies frequency-modulated Crz1 localization and is closely related to the class discussion of digital / bang-bang-like regulation.

6. Positive + negative feedback and tunable frequency

Delay oscillators and repressilators generate a period through the delay around a negative-feedback loop. Their different components typically oscillate **out of phase**. Because the frequency depends on the loop delay and the same kinetic parameters also affect the oscillation amplitude, it is difficult to change frequency without also changing amplitude.

Many biological oscillators use an additional motif: **positive feedback coupled to negative feedback**. These oscillations are typically more **spiky**, and the coupled components can oscillate approximately **in phase**.

A key idea is **time-scale separation**. One timescale mainly determines the switching behavior and the pulse amplitude, whereas another timescale mainly determines the interval between pulses. Changing a parameter that controls the latter can therefore change the oscillation frequency while leaving the amplitude approximately unchanged.

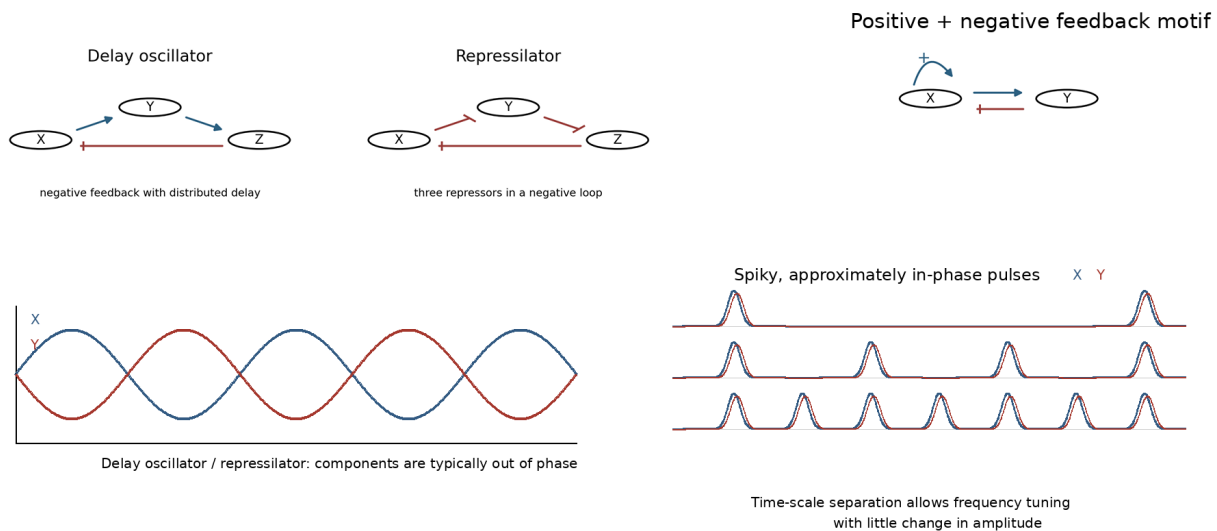


Figure 6. Left: delay oscillator and repressilator, where the coupled components are typically out of phase. Right: positive + negative feedback, where the components can oscillate approximately in phase. Based on Uri Alon Lecture 9 / Chapter 6.

The next section uses the **cell-cycle oscillator** as a concrete example to show how positive + negative feedback together with time-scale separation produces this spiky, tunable-frequency behavior.

7. The CDK1–APC/C cell-cycle oscillator

The cell-cycle example combines positive and negative feedback.

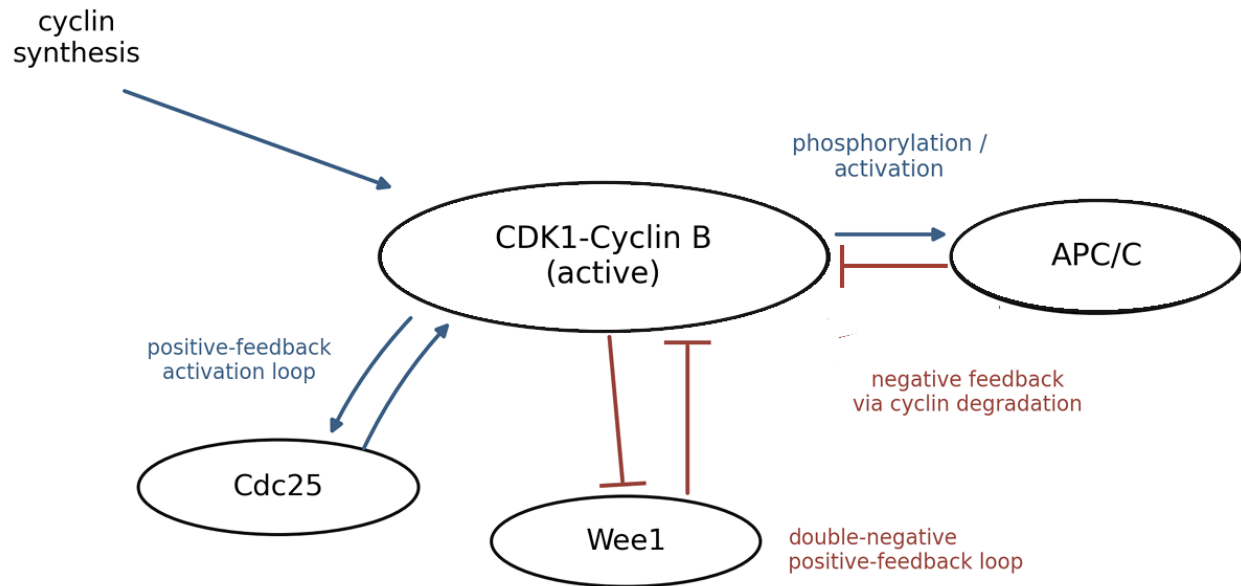


Figure 7. CDK1–APC/C relaxation oscillator. CDK1 activates APC/C, which promotes cyclin degradation, and is reinforced by the Cdc25/Wee1 positive-feedback arms. Redrawn from Pomerening et al. (2003) and Yang & Ferrell (2013).

The logic is:

cyclin accumulation $\rightarrow$ CDK1 activation $\rightarrow$ APC/C activation
 $\rightarrow$ cyclin degradation $\rightarrow$ CDK1 inactivation.

The positive-feedback arms make CDK1 behave as a sharp switch. The slower accumulation and removal of cyclin then repeatedly drive the system across the switching thresholds.

Because changing the cyclin synthesis rate mainly changes how quickly the threshold is reached, the oscillator can change its **frequency** while keeping the CDK1 pulse amplitude approximately fixed. In this sense, the cell-cycle clock behaves like an FM system.

8. Hysteresis turns a bistable switch into a relaxation oscillator

A bistable CDK1 switch has two different switching thresholds. The state therefore depends not only on the current cyclin level, but also on whether cyclin is increasing or decreasing.

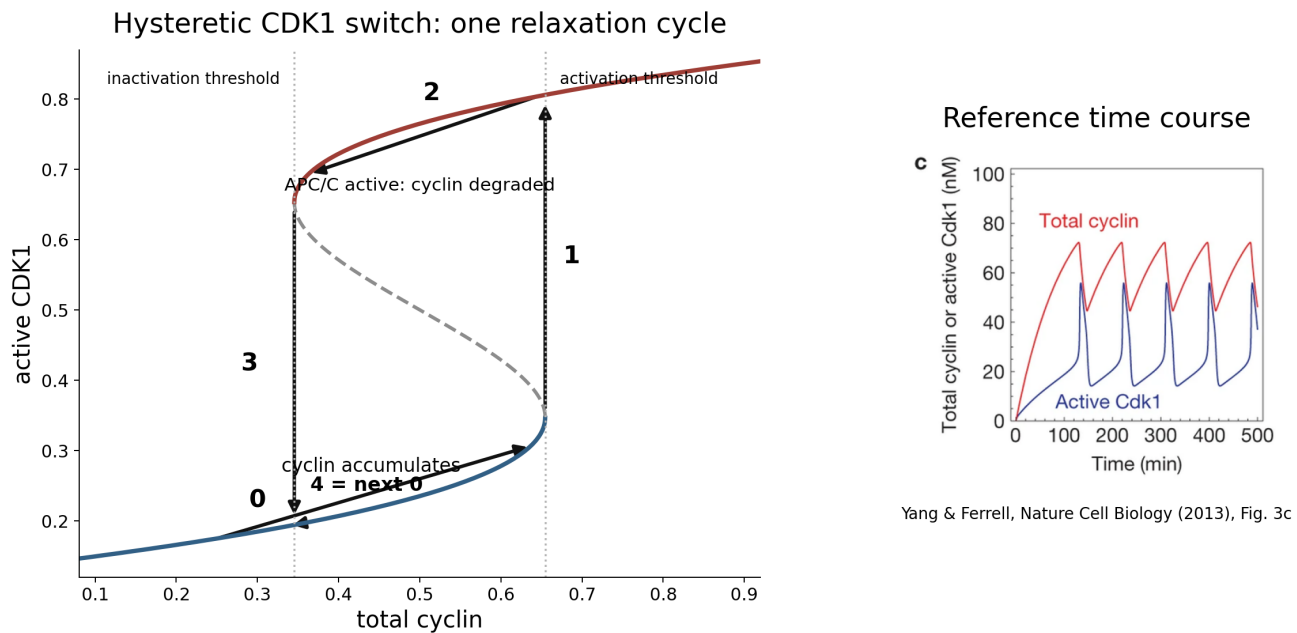


Figure 8. Left: corrected schematic of one hysteretic relaxation cycle. Right: the model time course from Yang & Ferrell (2013), Fig. 3c, showing saw-toothed total cyclin and sharp active-CDK1 pulses. Source: Yang, Q. & Ferrell, J. E. Jr, *Nature Cell Biology* **15**, 519–525 (2013), Fig. 3.

The cycle is read as follows:

1. **Stage 0:** CDK1 is low. APC/C is only weakly activated, cyclin degradation is low, and cyclin accumulates. The system moves to the right along the low-CDK1 stable branch until it reaches the activation threshold.
2. **Stage 1:** at the upper cyclin threshold, the low-CDK1 stable state disappears in a bifurcation. CDK1 rapidly jumps from the low branch to the high branch.
3. **Stage 2:** high CDK1 activates APC/C. APC/C degrades cyclin, so total cyclin decreases and the system moves left along the high-CDK1 branch.
4. **Stage 3:** at the lower cyclin threshold, the high-CDK1 stable state disappears in a second bifurcation. CDK1 rapidly jumps from the high branch back to the low branch.
5. **Stage 4:** with CDK1 low, APC/C is phosphorylated and activated less strongly. Cyclin degradation drops below cyclin synthesis, so cyclin begins to accumulate again. This is the beginning of the next cycle, so **stage 4 is the next stage 0**.

The important role of hysteresis is that after CDK1 switches on, it remains high while cyclin is being degraded; it does not immediately switch back off. This allows the slow cyclin variable to carry the system from one switching threshold to the other and produces a relaxation cycle.

9. Why use two positive-feedback loops? Robust bistability

The CDK1 switch uses two reinforcing feedback arms: CDK1 activates Cdc25, which activates CDK1, and CDK1 inhibits Wee1, which normally inhibits CDK1. The question is why both

arms are retained when either loop can, in principle, contribute to bistability.

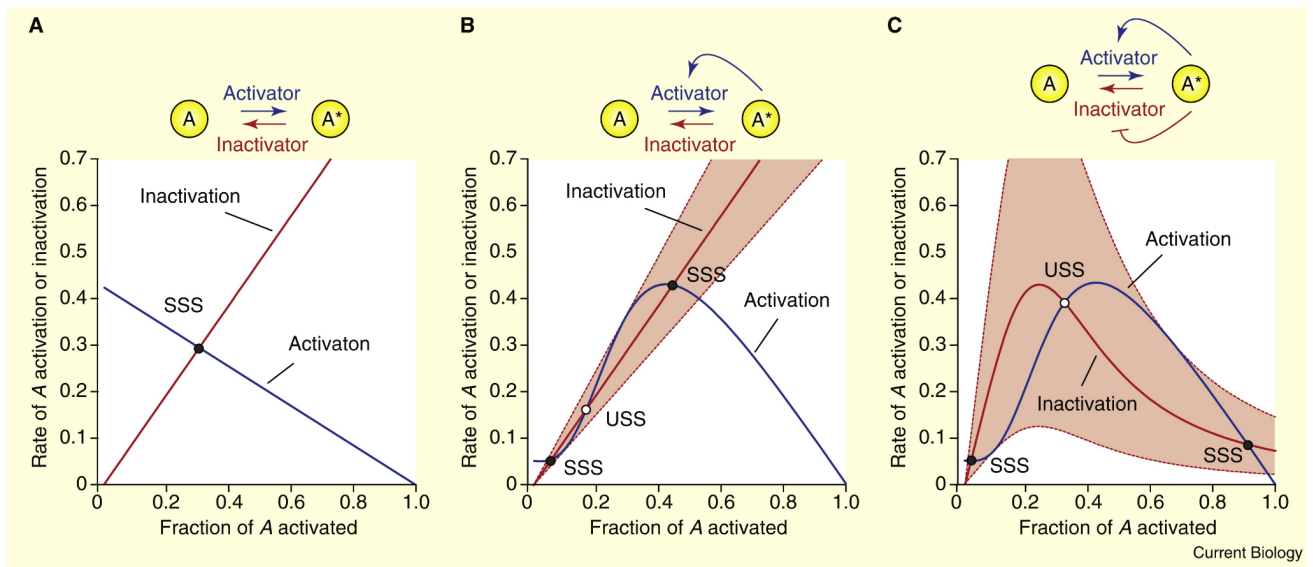


Figure 9. Reciprocal feedback regulation of opposing enzymes broadens the parameter range that supports bistability. Black points are stable steady states (SSS); open points are unstable steady states (USS). Source: Ferrell, J. E. Jr, “Feedback regulation of opposing enzymes generates robust, all-or-none bistable responses,” *Current Biology* **18**, R244–R245 (2008).

The rate plots make the main point visually. With no feedback, there is a single stable intersection. With feedback on only one side, three intersections—and therefore bistability—can occur, but only over a more restricted parameter range. When the activator and inactivator are regulated reciprocally, the region supporting three intersections becomes wider. The two feedback arms therefore make the bistable switch **more robust to changes in kinetic parameters**.

10. Take-home points

- An autonomous one-dimensional system cannot generate a periodic trajectory.
- A two-node nonlinear negative-feedback system shows damped oscillations when $(\alpha_1 - \alpha_2)^2 < 4\beta_1\beta_2$; similar timescales and stronger feedback favor this regime. It cannot support a sustained limit cycle.
- Noise can repeatedly excite a damped p53–Mdm2 oscillator, producing pulses with a comparatively precise frequency but noisy amplitude.
- Oscillatory signals can encode input strength through amplitude modulation or frequency modulation, but these modes have different consequences for downstream readout.
- Coupled positive and negative feedback, together with time-scale separation, can produce spiky oscillations whose frequency is tunable with relatively little change in amplitude.
- In the CDK1–APC/C oscillator, hysteresis converts cyclin accumulation and degradation into repeated switching between low- and high-CDK1 states, while reciprocal positive-feedback regulation makes the bistable switch more robust.

Appendix A. Terms to check after class

- AM — amplitude modulation
- FM — frequency modulation
- Positive feedback
- Negative feedback
- Cell cycle
- Hysteresis
- Relaxation oscillator
- Tunable frequency
- Double / reciprocal feedback
- Rate plot
- Robust bistability
- Overdamping / overdamped response
- Underdamping / underdamped response
- Noise-induced oscillation
- Phase portrait
- Stable spiral
- Stable node
- Fixed point
- Amplitude
- Feedback gain β
- Timescale τ
- Delay oscillator
- Repressilator
- Bistability

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9. Yang, Q. & Ferrell, J. E. Jr. The Cdk1–APC/C cell cycle oscillator circuit functions as a time-delayed, ultrasensitive switch. *Nature Cell Biology* **15**, 519–525 (2013).
10. Yissachar, N. et al. Dynamic response diversity of NFAT isoforms in individual living cells. *Molecular Cell* **49**, 322–330 (2013). **Optional reading; not covered in class.**
11. Cai, L., Dalal, C. K. & Elowitz, M. B. Frequency-modulated nuclear localization bursts coordinate gene regulation. *Nature* **455**, 485–490 (2008). **Optional reading; not covered in class.**